

**MA.912.AR.9** Write and solve a system of two- and three-variable equations and inequalities that describe quantities or relationships.

### MA.912.AR.9.1

#### **Benchmark**

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**MA.912.AR.9.1** Given a mathematical or real-world context, write and solve a system of two-variable linear equations algebraically or graphically.

#### Benchmark Clarifications:

*Clarification 1:* Within this benchmark, the expectation is to solve systems using elimination, substitution and graphing.

*Clarification 2:* Within the Algebra I course, the system is limited to two equations.

#### **Connecting Benchmarks/Horizontal Alignment**

- MA.912.AR.2.1, MA.912.AR.2.2, MA.912.AR.2.3, MA.912.AR.2.4

#### **Terms from the K-12 Glossary**

- Linear Equation

#### **Vertical Alignment**

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##### **Previous Benchmarks**

- MA.8.AR.4

##### **Next Benchmarks**

- MA.912.NSO.4.2
- MA.912.AR.9.2, MA.912.AR.9.3, MA.912.AR.9.9

#### **Purpose and Instructional Strategies**

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In grade 8, students determined whether a system of linear equations had one solution, no solution or infinitely many solutions and solved such systems graphically. In Algebra I, students solve systems of linear equations in two variables algebraically and graphically. In later courses, students will solve systems of linear equations in three variables and systems of nonlinear equations in two variables.

- For students to have full understanding of systems, instruction should include MA.912.AR.9.4 and MA.912.AR.9.6. Equations and inequalities and their constraints are all related and the connections between them should be reinforced throughout instruction.
- Instruction allows students to solve using any method (substitution, elimination or graphing) but recognizing that one method may be more efficient than another (*MTR.3.1*).
  - If both equations are presented in slope-intercept form, then either graphing or substitution may be most efficient.
  - If one equation is given in slope-intercept form or solved for  $x$ , then substitution may be easiest.
  - If both equations are given in standard form, then elimination, or linear combination, may be most efficient.

Consider presenting a system that favors one of these methods and having students divide into three groups to solve them using different methods. Have students share their work and discuss which method was more efficient than the others (*MTR.3.1*, *MTR.4.1*).

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- Include cases where students must interpret solutions to systems of equations.
  - Instruction includes the use of various forms of linear equations.
    - Standard Form  
Can be described by the equation  $Ax + By = C$ , where  $A$ ,  $B$  and  $C$  are any rational number.
    - Slope-Intercept Form  
Can be described by the equation  $y = mx + b$ , where  $m$  is the slope and  $b$  is the  $y$ -intercept.
    - Point-Slope Form  
Can be described by the equation  $y - y_1 = m(x - x_1)$ , where  $(x_1, y_1)$  are a point on the line and  $m$  is the slope of the line.
  - When introducing the elimination method, students may express confusion when considering adding equations together. Historically, students have used the properties of equality to create equivalent equations to solve for a variable of interest. In most of these efforts, operations performed on both sides of the original equation have been identical. With the introduction of the elimination method, students can now see that operations performed on each side of an equation must be *equivalent* (not necessarily *identical*) for the property to hold. Guide students to explore forming equivalent equations with simpler equations by adding or subtracting equivalent values. Lead them to see that the new equations they generate have the same solutions. Have them discuss why the method works: equations are simply pairs of equivalent expressions, which is why they can be added/subtracted with each other.

### **Common Misconceptions or Errors**

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- Students may not understand linear systems of equations can only have more than one solution if there are infinitely many solutions.
- Students may not understand linear systems of equations can have no solution.
- Students may have difficulty making connections between graphic and algebraic representations of systems of equations.
- Students may have difficulty choosing the best method of finding the solution to a system of equations.
- Students may have difficulty translating word problems into systems of equations and inequalities.
- Students using the elimination method may alter the original equations in a way that creates like terms that can be subtracted. When subtracting across the two equations students may have difficulty remembering to apply the subtraction to the remaining terms and constants.

## Strategies to Support Tiered Instruction

- Instruction includes opportunities to use graphing software to visualize the possible solutions for a system of equations. Systems of equations only produce three different types of solutions: one solution, infinite solutions, and no solutions. Each type of system can be graphed for analysis of each type of solution set.
- Teacher models through a think-aloud how a system of equations can have no solutions.
  - For example, “I can algebraically solve a system with no solutions. The solution will reveal that the left and right sides of the equation cannot be equal, causing a no solution set. In addition, if I rearrange both equations to the slope-intercept form, the equations will have the same slope. I can utilize my knowledge of parallel lines to understand that the system cannot have any solutions.”
- Teacher provides step-by-step process for solving systems.
  - For example, when solving the system below, students can use the method of elimination.

$$2x + 4y = -10$$

$$3x + 5y = 8$$

If the student chooses to eliminate the  $y$ -variable, they can multiply the first equation by 5 and the second by 4 so that both coefficients of  $y$  are 20.

$$5(2x + 4y = -10) \text{ to } 10x + 20y = -50$$

$$4(3x + 5y = 8) \text{ to } 12x + 20y = 32$$

The student either subtracts the two new equations, or creates additive inverses by multiplying one of the equations by  $-1$  (as shown) and then adds the equations.

$$-1(10x + 20y = -50) \text{ to } -10x - 20y = +50$$

$$-10x - 20y = +50$$

$$\underline{12x + 20y = 32}$$

$$2x = 82$$

$$x = 41$$

Once students determine one of the values ( $x$  in this case), then they can substitute this back into one of the given equations to find the other value ( $y$  in this case).

$$2(41) + 4y = -10$$

$$4y = -10 - 82$$

$$y = -23$$

## Instructional Tasks

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### Instructional Task 1 (MTR.3.1, MTR.4.1)

You and a friend go to Tacos Galore for lunch. You order three soft tacos and three burritos and your total bill is \$11.25. Your friend's bill is \$10.00 for four soft tacos and two burritos.

Part A. Write a system of two-variable linear equations to represent this situation.

Part B. Solve the system both algebraically and graphically to determine the cost of each burrito and each soft taco.

Part C. Is one method more efficient than the other? Why or why not?

### Instructional Task 2 (MTR.3.1, MTR.4.1)

Part A. Determine the solution to the system of linear equations below using your method of choice.

$$\begin{aligned}0.5x - 1.4y &= 5.8 \\ y &= -0.3x - \frac{1}{5}\end{aligned}$$

Part B. Discuss with a partner why you chose that method.

## Instructional Items

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### Instructional Item 1

Determine the exact solution of the system of linear equations below.

$$\begin{aligned}-\frac{1}{10}x + \frac{1}{2}y &= \frac{4}{5} \\ \frac{1}{7}x + \frac{1}{3}y &= -\frac{2}{21}\end{aligned}$$

### Instructional Item 2

Carla volunteered to make pies for a bake sale. She bought two pounds of apples and six pounds of peaches and spent \$19. After baking the pies, she decided they looked so good she would make more. She went back to the store and bought another pound of apples and five more pounds of peaches and spent \$15. Write a system of linear equations that describes her purchases, where  $a$  represents the cost per pound of the apples and  $p$  represents the cost per pound of the peaches.

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*\*The strategies, tasks and items included in the B1G-M are examples and should not be considered comprehensive.*